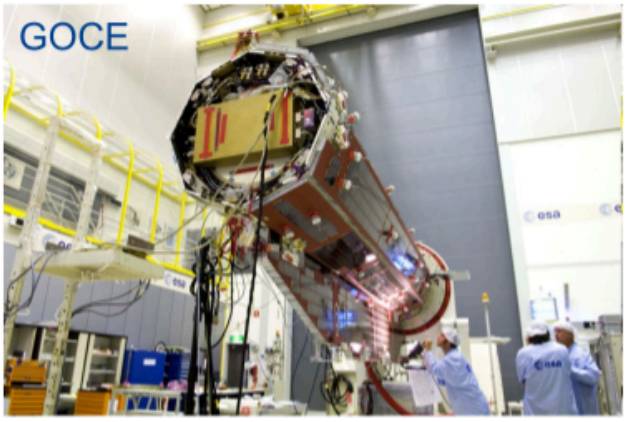
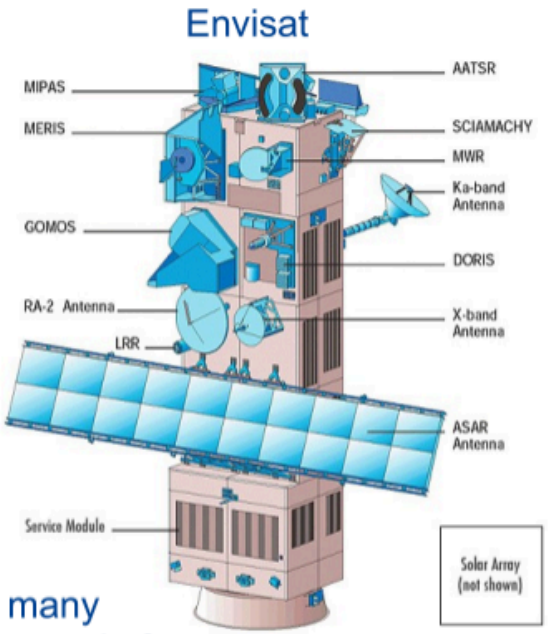
Lecture 5: Principles of Satellite Orbits

Satellites

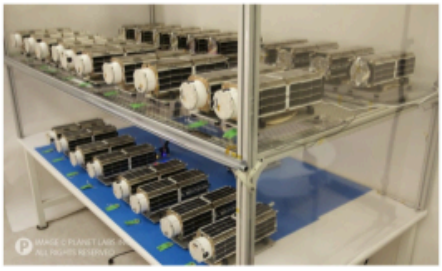
- Fundamental research
  - Astronomy, Astrophysics
  - Geodesy, gravity field, geophysics, oceanography
  - Climate, environment
- Earth Observation
  - Weather: Clouds, precipitation, radiation
  - Sea level, surface temperature/color/contamination/...
  - Sea Ice cover
  - Geophysics/Geology, Topography, land surface
  - Space weather, ionsphere, sun, magnetic field
- Telecommunication
- Navigation (inc. GNSS)



Dedicated: specialized, only 1 instrument



Multi-purpose: many instruments of one platform

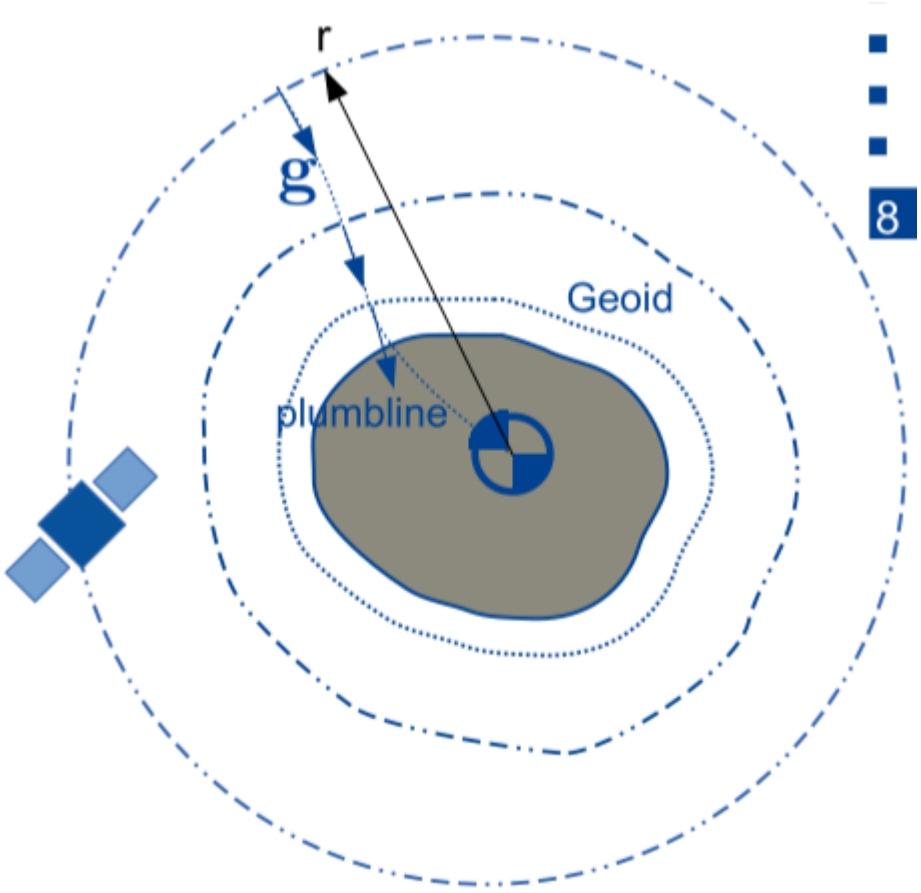


"Cubesats", "Nanosats": many small satellites, specialized, cost-effective
Starlink: > 8000 satellites for Internet access coverage

Orbits

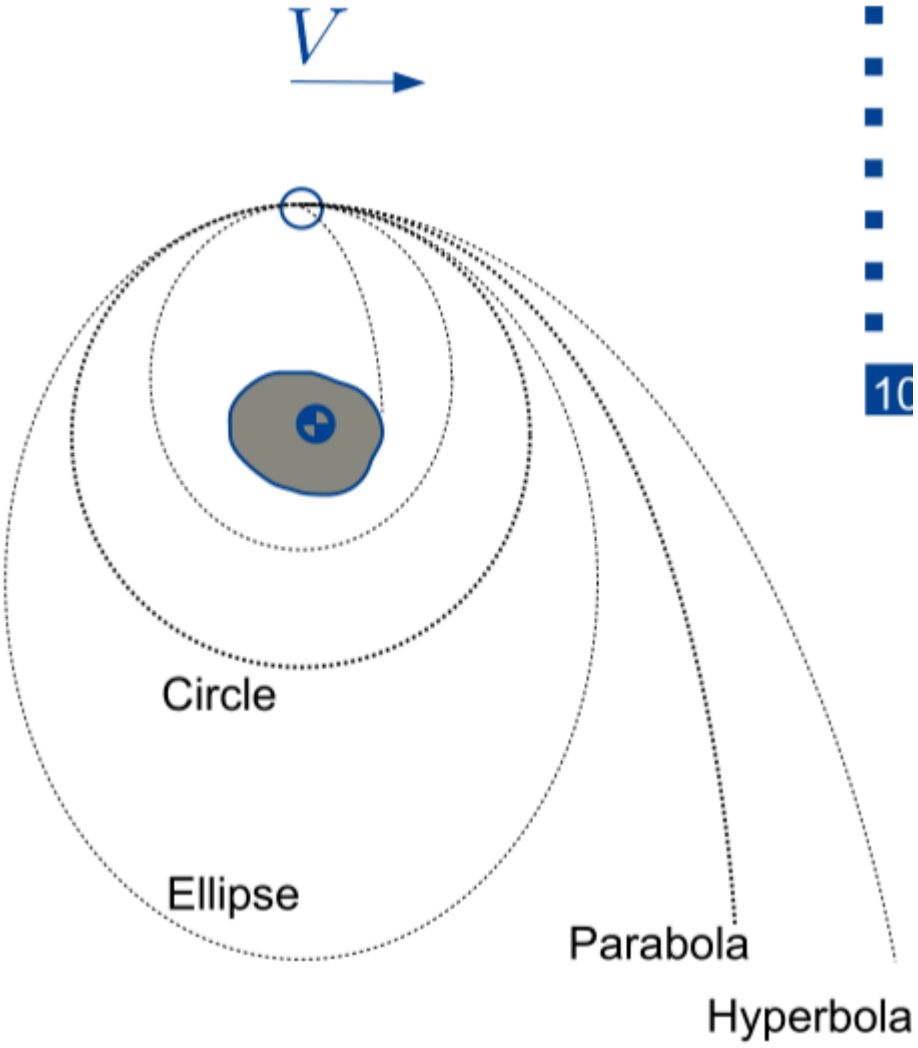
$$\lim_{r \to \infty} \boldsymbol{g} = \frac{GM\boldsymbol{r}}{|\boldsymbol{r}|^3}$$

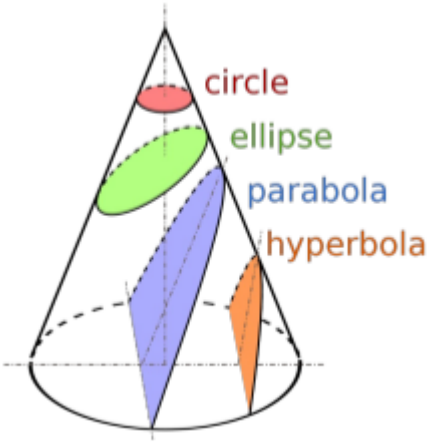
- $\boldsymbol{g}$ points towards center of mass
  - Only in special cases
  - From infinite distances
- Free-falling orbital objects do **not** move along equipotential surfaces
  - Coincidentally: point masses in combination with a circular orbit



Orbital paths

- Increase horizontal velocity, V
  - Orbit
  - Escape orbit



- Earth as a point mass
  - -> Kepler orbit
  - Kepler orbits are conical sections
  - 

Kepler Orbits

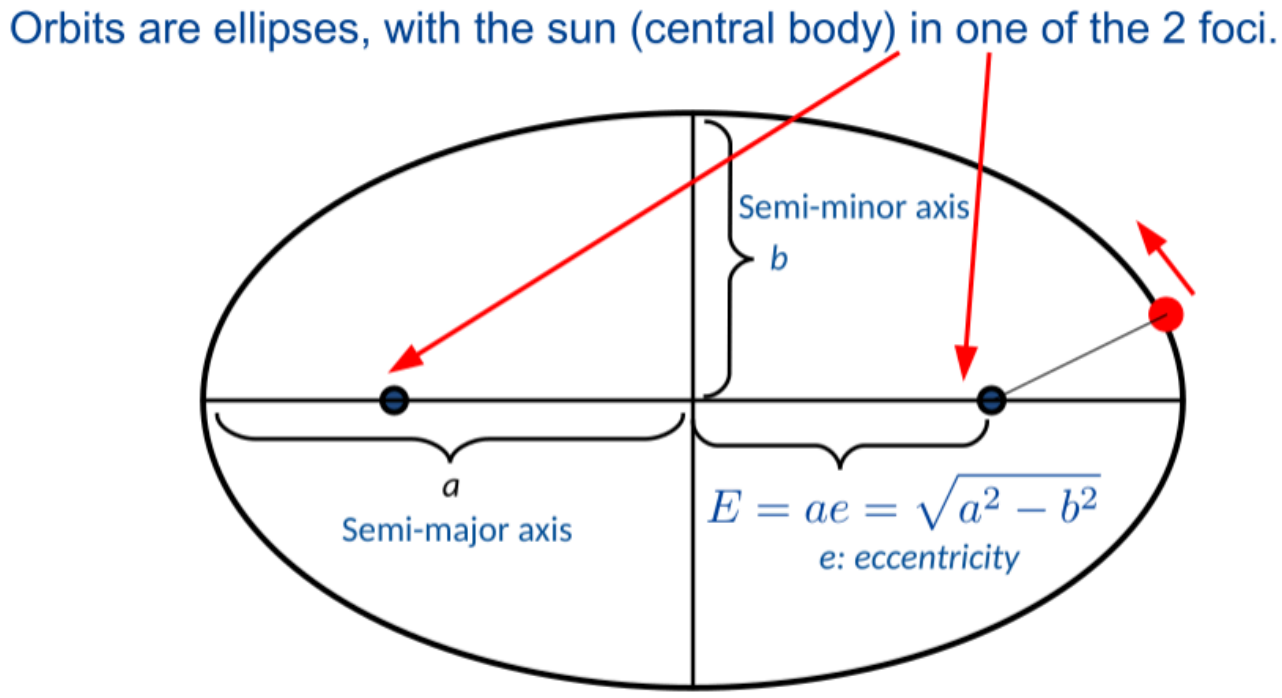
- Assumption: Earth is a point mass
- Satellite trajectory (Position and Time)
- -> Kepler elements

- **History**:
  - Tycho Brahe compiled comprehensive database of astronomical observations
  - Johannes Kepler explained these with 3 laws of planetary motion (assistant and follow up of Bahe in Prague)

Kepler Laws

1st Law - Central body

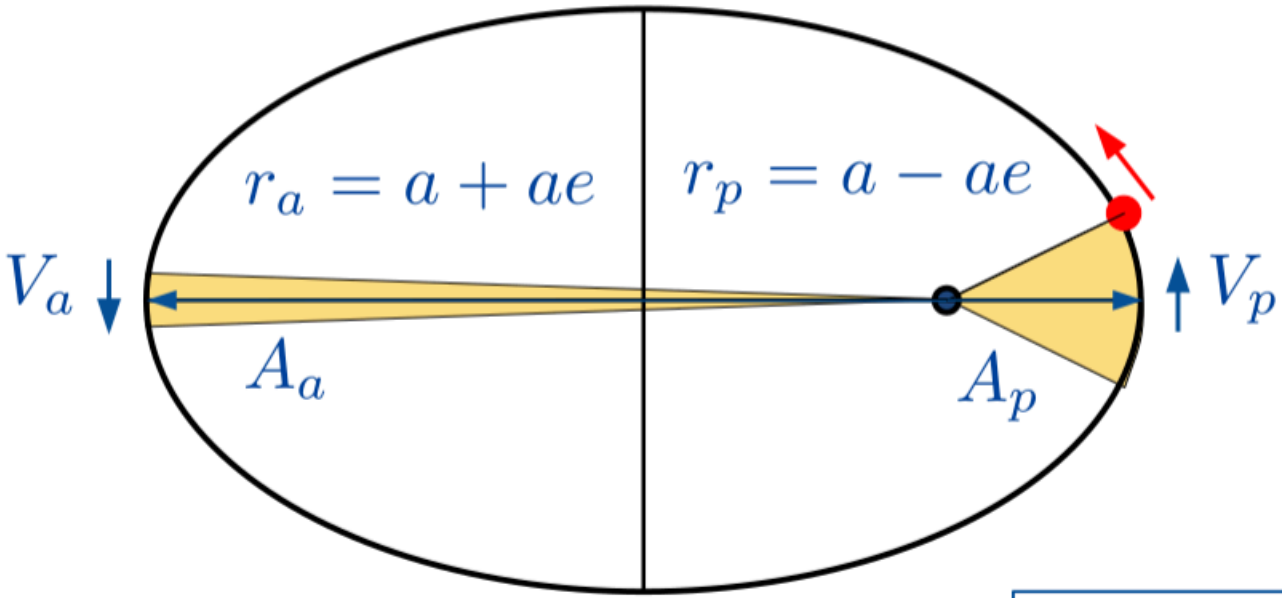
- Orbits are ellipses with sun (central body) in one of the foci



  - From lecture 1 → same math, different application

2nd Law - Equal times, Equal areas

- During equal intervals of time, the connecting line between central body and planet/objects sweeps equal areas

- 

  - p: perihelion/perigee
  - a: aphelion/apogee
  - Here:

$$V_a \leq V_p$$

-
  - Compute areas for small differences:
    - $A = \frac{1}{2}rh = \frac{1}{2}r\sin(\alpha)vdt$
    - With $\alpha = 90^\circ$: $A = \frac{1}{2}rvdt$
  - Apply Kepler's 2nd law
    - $A_a = dtV_a(a + ae)/2 = A_p = dtV_P(a - ae)/2$
    - $V_a = \frac{1-e}{1+e}V_p, \quad =\leq e \leq 1$
- Law is consequence of physics
  - Specific angular momentum **h** of an orbiting object (measure of swept area)

$$\boxed{\dot{\mathbf{v}} = -\frac{GM}{x^3}\mathbf{x}} \rightarrow \boxed{\mathbf{x} \times \dot{\mathbf{v}} = -\frac{GM}{x^3}\underbrace{\mathbf{x} \times \mathbf{x}}_{=0}} \rightarrow \boxed{\mathbf{x} \times \dot{\mathbf{v}} = 0}$$

Equation of motion

$$\boxed{\frac{d}{dt}(\mathbf{x} \times \mathbf{v}) = \mathbf{v} \times \mathbf{v} + \mathbf{x} \times \dot{\mathbf{v}} = \mathbf{x} \times \dot{\mathbf{v}}} \rightarrow \boxed{\frac{d}{dt}(\mathbf{x} \times \mathbf{v}) = 0}$$

Auxiliary calculation

$$\boxed{\mathbf{x} \times \mathbf{v} = \mathbf{h} = const}$$

  -

$$\boldsymbol{x} \times \boldsymbol{v} = \boldsymbol{h} = const$$

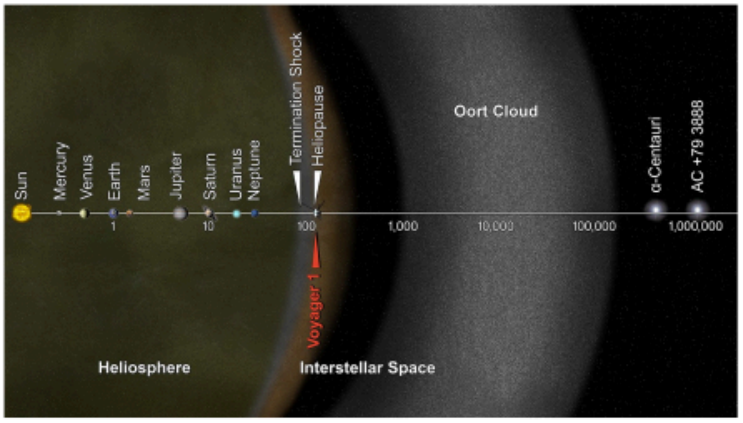
1. Specific angular momentum is constant
   - $m\dot{\boldsymbol{h}} = M_{extern} = 0$
   - No external mass moments
   - Areas remain constant in time
2. Position vector and velocity vectors are perpendicular to $\boldsymbol{h}$ -> constant orbital plane

3rd Law - Orbital period

- Square of the orbital period is proportional to the cube of the semi major axis of the ellipse

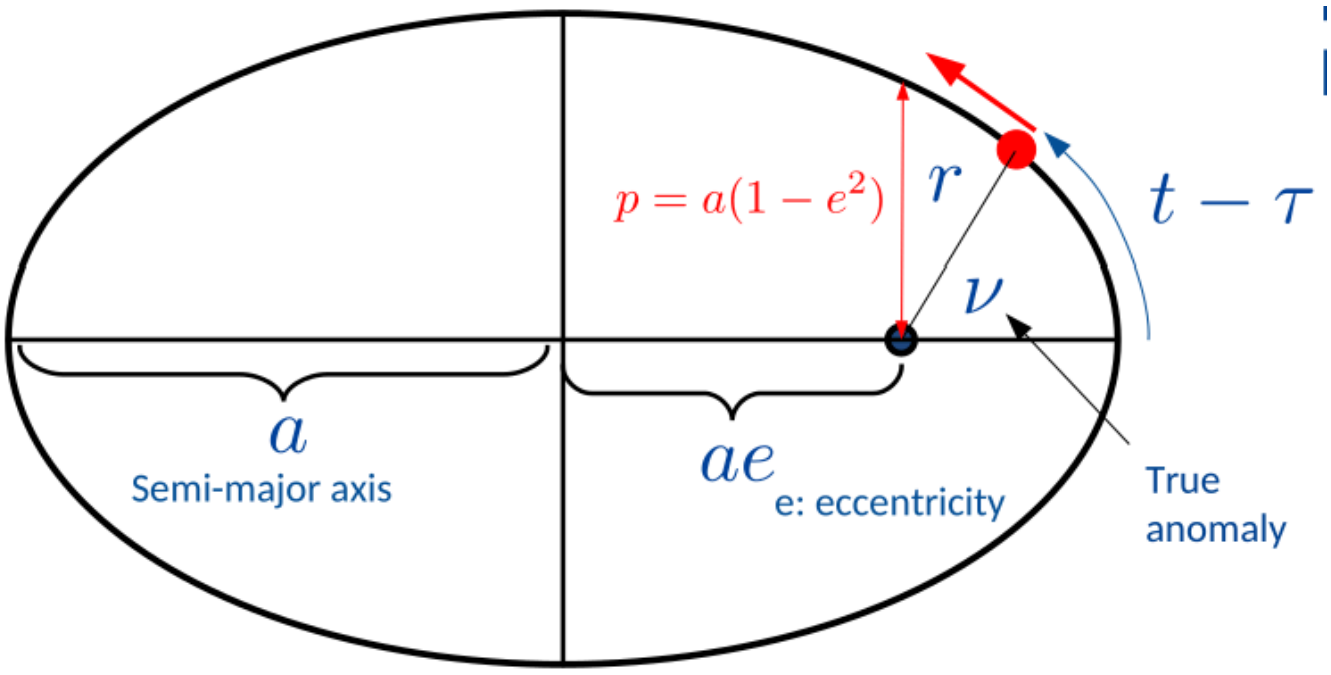
$$T^2 = 4\pi^2 \frac{a^3}{GM}$$

$$\frac{T_s}{T_e} = \sqrt{\frac{a_s^3}{a_e^3}} \approx 32$$



In Plane Kepler elements

- Describe size and shape of orbit (2 parameters) a, e
- Set time of periapsis passing (1 parameter) τ



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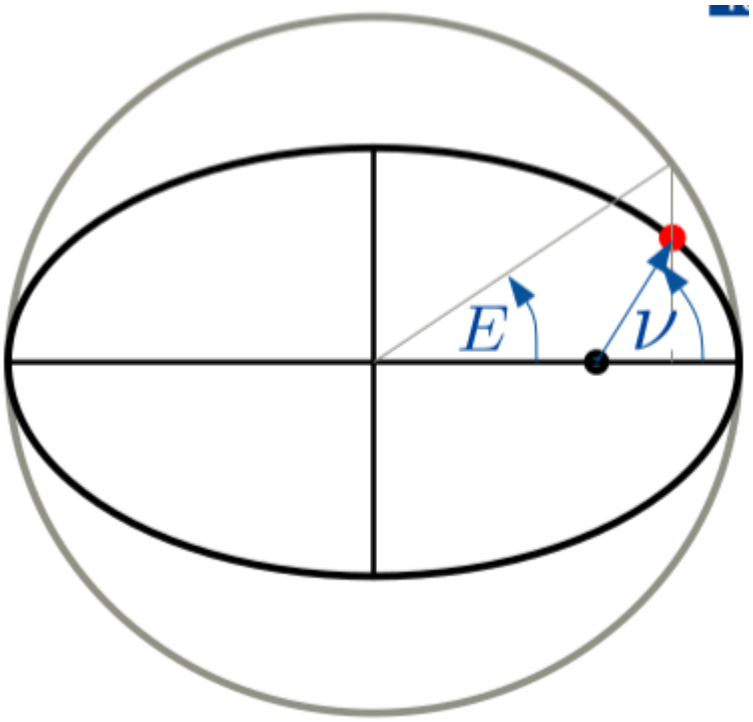
In Plane Geometry: Angular anomalies

- ν - True anomaly
- E Eccentric anomaly
- Mean anomaly $M = n(t - \tau)$
  - mean angular velocity $n = \frac{2\pi}{T} = \sqrt{\frac{GM}{a^3}}$
- Kepler Equation
  - $M = E - e\sin(E)$

$$\cos\nu = \frac{\cos E - e}{1 - e\cos E} \quad or \quad \sin\nu = \frac{\sqrt{1-e^2}\sin E}{1 - e\cos E}$$

$t \to M \to E \to \nu$



In plane orbital geometry: Position & Velocity

- ν - True anomaly
- E Eccentric anomaly
- Mean anomaly M
- $p = a(1 - e^2)$

Position

$$\mathbf{r} = \begin{bmatrix} r\cos\nu \\ r\sin\nu \\ 0 \end{bmatrix}$$

Velocity

$$\dot{\mathbf{r}} = \frac{GM}{C}\begin{bmatrix} -\sin\nu \\ \cos\nu + e \\ 0 \end{bmatrix}$$

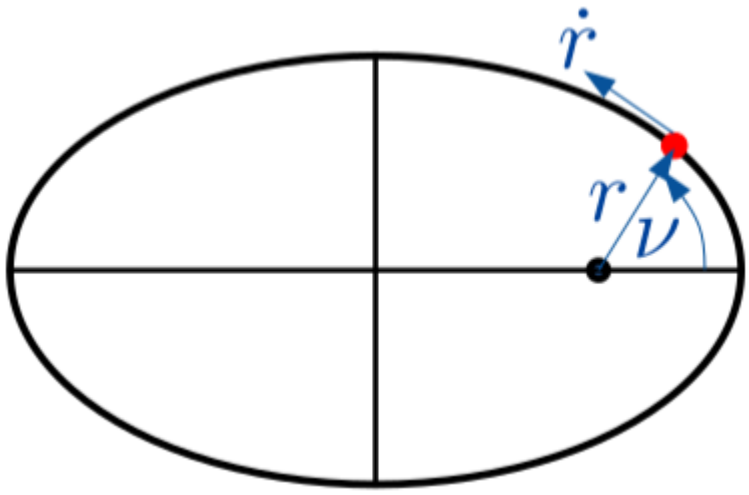
$$C = \sqrt{GMp}$$

$$r = \frac{p}{1 + e\cos\nu}$$

Ellipse equation

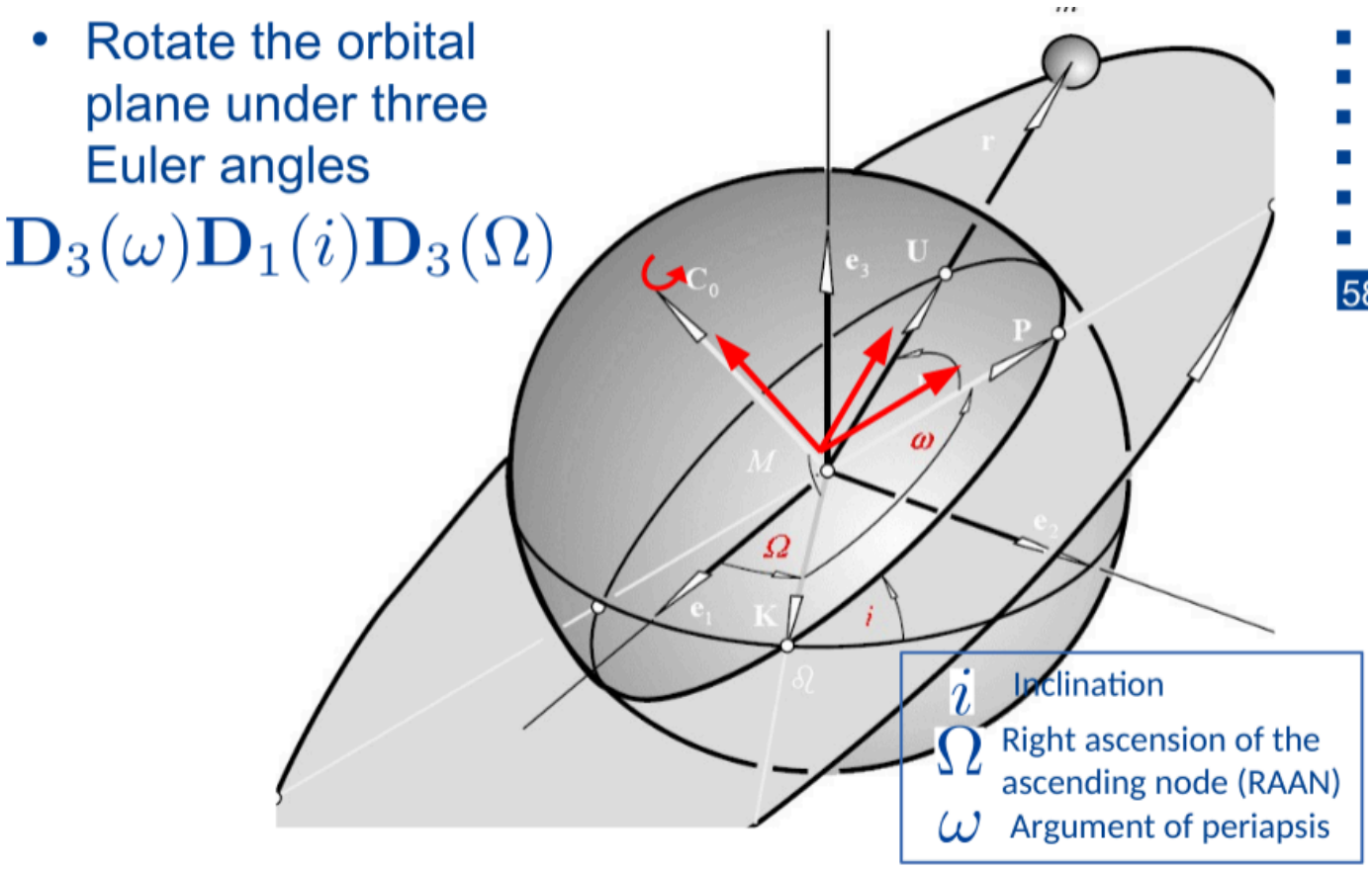
Solves **in-plane** problem:

$e, a, t \rightarrow \nu \rightarrow \boldsymbol{r}, \dot{\boldsymbol{r}}$



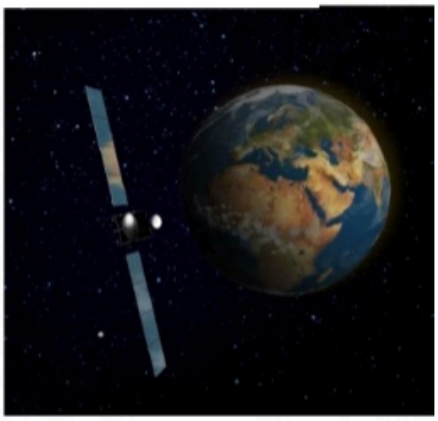
Ellipse in 3D -> 3 more Kepler parameters (angles) for rotation

Orientational Kepler elements



- Fix shape, and orbital period
  - a Semi-Major Axis
  - e Eccentricity
- Reference in time
  - τ Time of Periapsis passing
    - Sometimes mean anomaly or true anomaly
- Orient orbital plane in celestial reference system
  - i Inclination
  - Ω Right ascension of the ascending node (RAAN)
  - ω Argument of periapsis
- Using 6 Kepler elements versus 6 cartesian components (position & velocity)

Television satellite



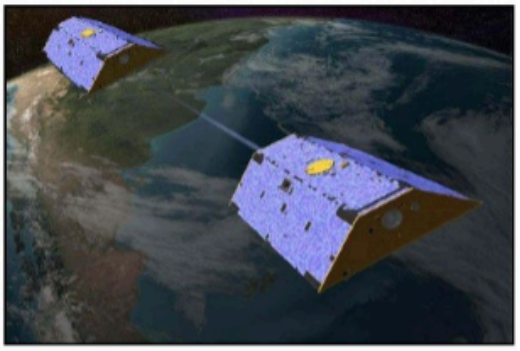
$a = 42.164$ km

$i = 0$

$e = 0$

$\left.\begin{array}{l}\Omega \\ \tau \\ \varpi\end{array}\right\}$ Not defined

Earth observation satellite



Example GRACE / GRACE-FO

$a \approx 6828$ km (Earth radius + 450 km)

$i = 90°$ (polar orbit)

$e \approx 0$ (circular orbit)

Relation of Earth rotation to orbital period should not be an integer
→ no repeat orbit

Orbital Energy

- Energy of an orbiting object consists of
  - Kinetic energy
  - Potential energy
- Sum of 2 remains constant and is only dependent on semi-major axis

$$|\mathbf{v}|^2 = GM\left(\frac{2}{r} - \frac{1}{a}\right)$$

$$\underbrace{\frac{1}{2}mv^2}_{\text{Kinetic energy}} - \underbrace{m\frac{GM}{r}}_{\text{Potential energy}} = \underbrace{-m\frac{GM}{2a}}_{\text{Total energy}}$$

Orbital energy depends only on the semi-major axis

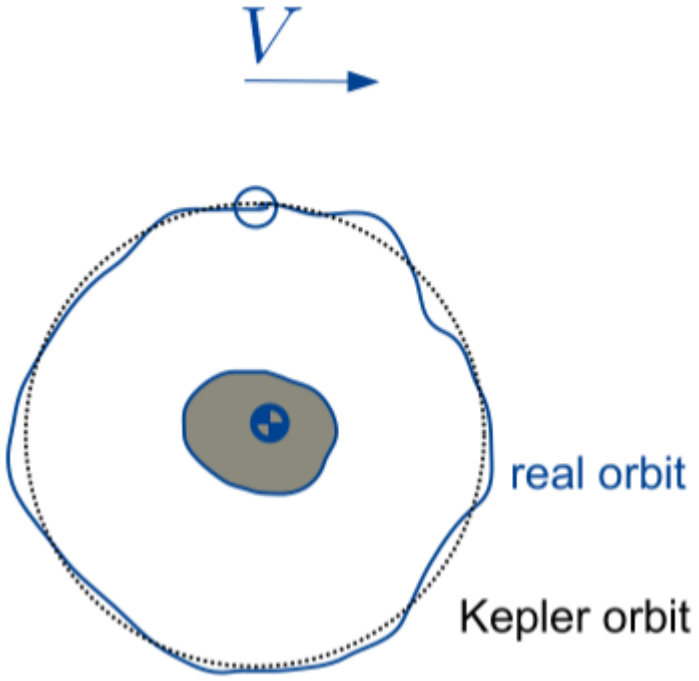
- Variety of forces act on satellites

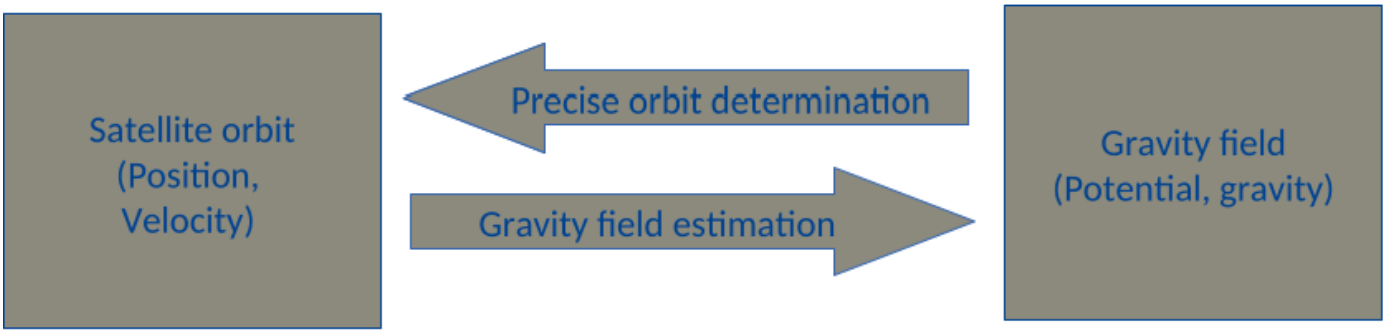
Table 6.1. Magnitude of perturbing accelerations (in m/sec^2 after [22])

source	STARLETTE	AJISAI	LAGEOS	GPS
a	7337km	7869km	12266km	26559km
$\frac{\text{area}}{\text{mass}}$	$9.6 \cdot 10^{-4}\text{m}^2/\text{kg}$	$5.3 \cdot 10^{-3}\text{m}^2/\text{kg}$	$6.9 \cdot 10^{-4}\text{m}^2/\text{kg}$	$2.0 \cdot 10^{-2}\text{m}^2/\text{kg}$
Kepler	7.4	6.4	2.6	0.6
J_2	$8 \cdot 10^{-3}$	$6 \cdot 10^{-3}$	$2 \cdot 10^{-3}$	$5 \cdot 10^{-5}$
other c_{nm}, s_{nm}	$1 \cdot 10^{-4}$	$9 \cdot 10^{-5}$	$5 \cdot 10^{-6}$	$3 \cdot 10^{-7}$
Sun, Moon	$1 \cdot 10^{-6}$	$1 \ldots 2 \cdot 10^{-6}$	$2 \cdot 10^{-6}$	$5 \cdot 10^{-6}$
Earth tides	$2 \cdot 10^{-7}$	$1 \ldots 2 \cdot 10^{-7}$	$3 \cdot 10^{-8}$	$1 \cdot 10^{-9}$
ocean tides	$3 \cdot 10^{-8}$	$2 \cdot 10^{-8}$	$2 \cdot 10^{-9}$	$1 \cdot 10^{-10}$
atmosph. drag	$1 \ldots 2 \cdot 10^{-10}$	$1 \ldots 2 \cdot 10^{-10}$	$3 \cdot 10^{-12}$ (empir.)	0
solar radiation	$5 \cdot 10^{-9}$	$5 \cdot 10^{-8}$	$4 \cdot 10^{-9}$	$1 \cdot 10^{-7}$
Earth albedo	$5 \cdot 10^{-10}$	$8 \cdot 10^{-9}$	$7 \cdot 10^{-11}$	$1 \cdot 10^{-9}$

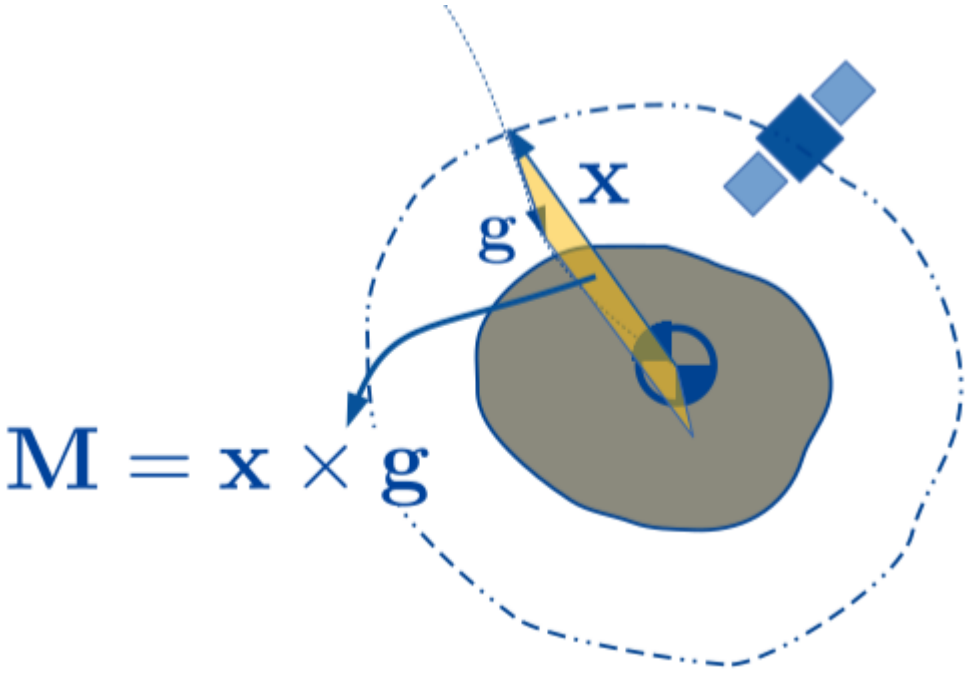
Orbital paths

- Anomalies from perfect Kepler orbit contain gravity field information
- Anomalies decreases for higher orbital altitudes
- Largest gravity anomaly: Earth's flattening





Orbital drift due to Earth's flattening



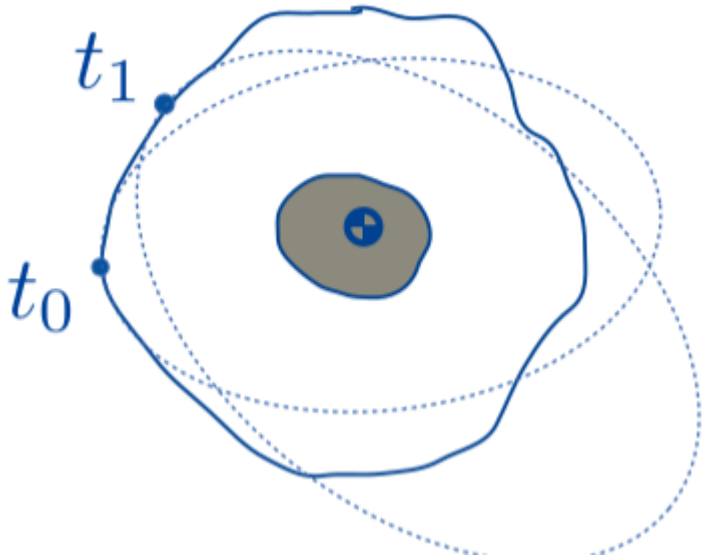
$$\boldsymbol{h} = \boldsymbol{x} \times \boldsymbol{v}$$
$$m\dot{\boldsymbol{h}} = \boldsymbol{M}_{extern} \neq 0$$

- Ellipsoidal bulge (J_2, C_{20}) of Earth exerts a **moment** on the orbit
  - **h** changes over time
  - Drift of the orbital plane

Strength of effect & affected Kepler elements?

Osculating Orbit

- Idea: For every point t, fit Kepler orbit through actual orbit
- The acceleration at any point in time is represented by the gravity induced by a point mass



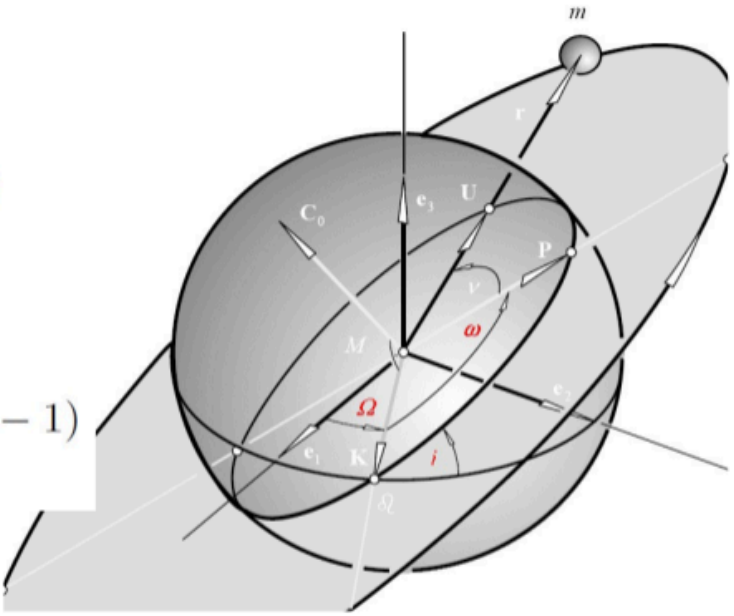
$$\boldsymbol{x}(t) = \tilde{\boldsymbol{x}}(\boldsymbol{c}(t), t)$$
$$\frac{d\boldsymbol{x}(t)}{dt} = \tilde{\boldsymbol{v}}(\boldsymbol{c}(t), t)$$

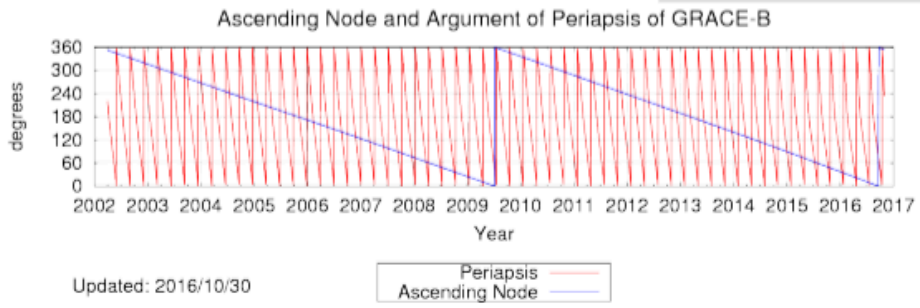
$$\frac{d^2\boldsymbol{x}}{dt^2} = -\frac{GM}{|\boldsymbol{x}|^3}\boldsymbol{x} + \boldsymbol{x}(\boldsymbol{x}, \dot{\boldsymbol{x}}, t)$$

- **Position** and **Velocity** computed from **Kepler orbit**
- But **acceleration** affected by **perturbing forces** as well
- Needed: Vector of time/ variable Kepler elements **c**(t)
  - $(\boldsymbol{c}(t) = [a(t), e(t), \ldots, \omega(t)])$
- After much math: (considering only gravity and linearisation)
- Only consider effects which do not average out
- Only use degree 2 order 0 term of gravity field

$J_2 = -\sqrt{5}\,\overline{C}_{20}$

$$\frac{d}{dt}a = \frac{d}{dt}e = \frac{d}{dt}i = 0$$
$$\frac{d}{dt}\omega = \frac{3R^2nJ_2}{4(1-e^2)^2a^2}(5\cos^2 i - 1)$$
$$\frac{d}{dt}\Omega = -\frac{3R^2nJ_2}{2(1-e^2)^2a^2}\cos i$$
$$\frac{d}{dt}M = n + \frac{3R^2nJ_2}{4(1-e^2)^{\frac{3}{2}}a^2}(3\cos^2 i - 1)$$





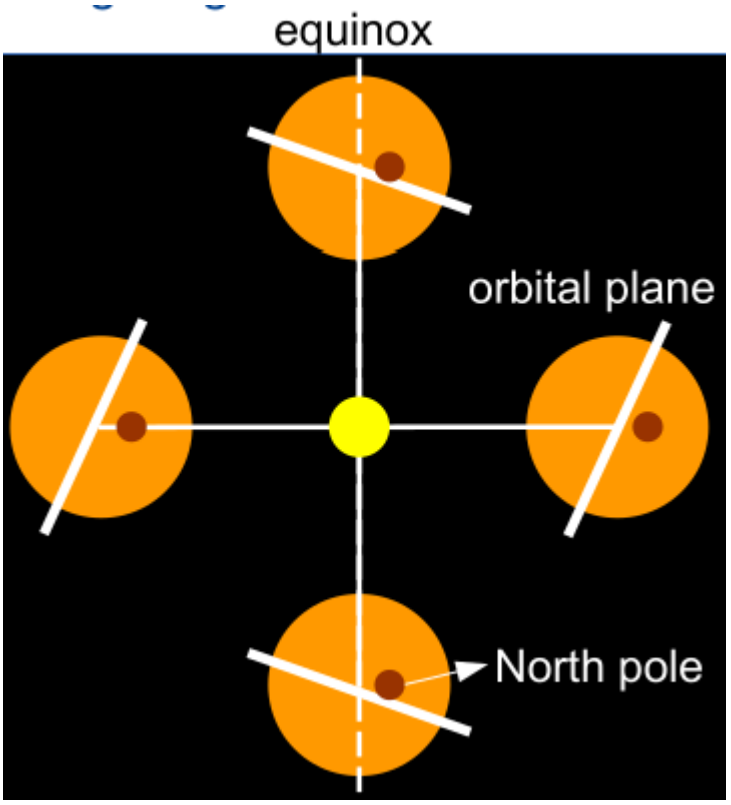
Special Orbit Types

- Some satellite orbit designs take advantage of flattening effect
  - Sun-synchronous orbits
  - Repeat orbit (radar altimetry)

Sun-Synchronous Orbits

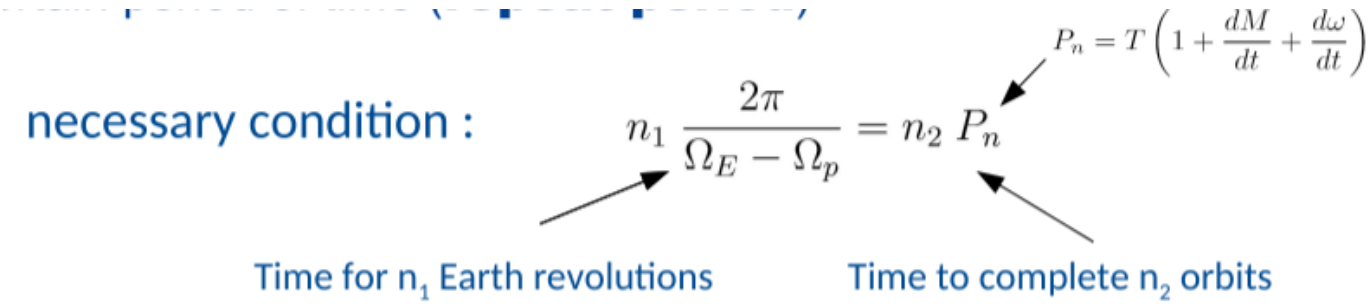
- Orbital plane always same angle w.r.t. sun
- e.g. no eclipse, predictable lighting conditions

- $\frac{d\Omega}{dt} = \frac{2\pi}{year}$
- Satellite passes same latitude (ascending or descending) at the same local solar time
    - i.e. same sun illumination angle



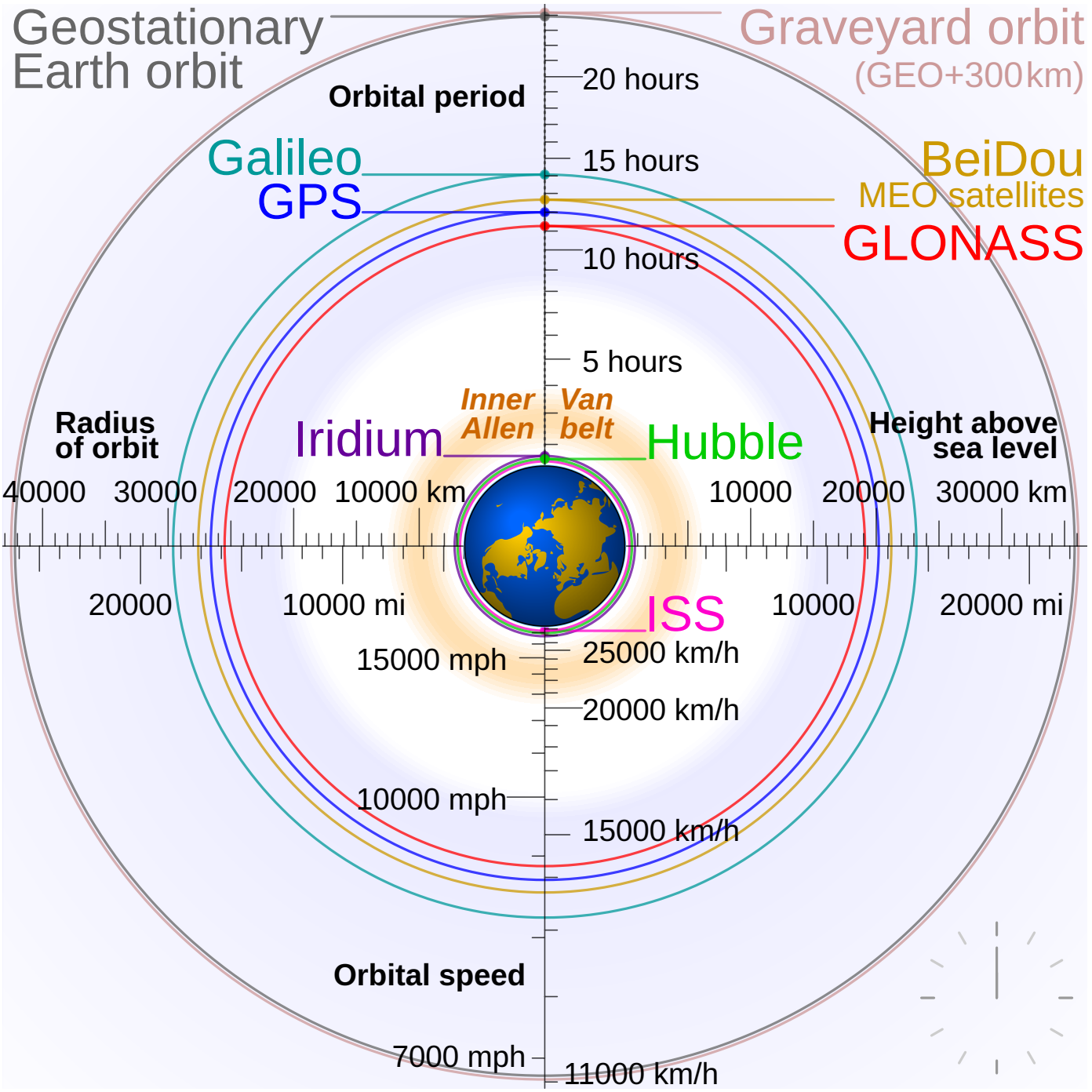
Repeat orbit

- Satellite repeats it's groundtrack pattern after certain **repeat period**
    - A $cycle$ is one full completion of the repeat period



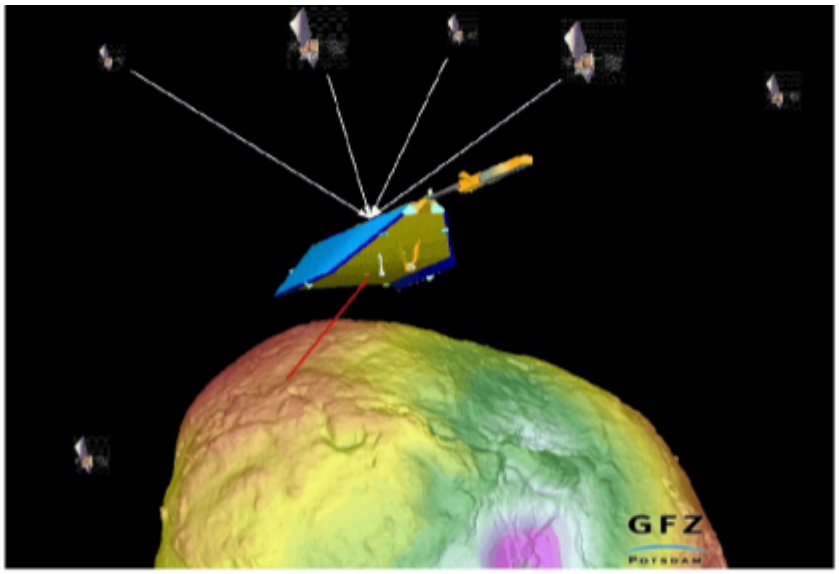
- $P_n(\Omega_E - \Omega_p) = 2\pi\frac{n_1}{n_2}$ "n_1-day repeat orbit"
- $\frac{P_n}{P'_E} = \frac{n_1}{n_2}$ for **sun-synchronous orbit** (P'_E = 1 solar day)

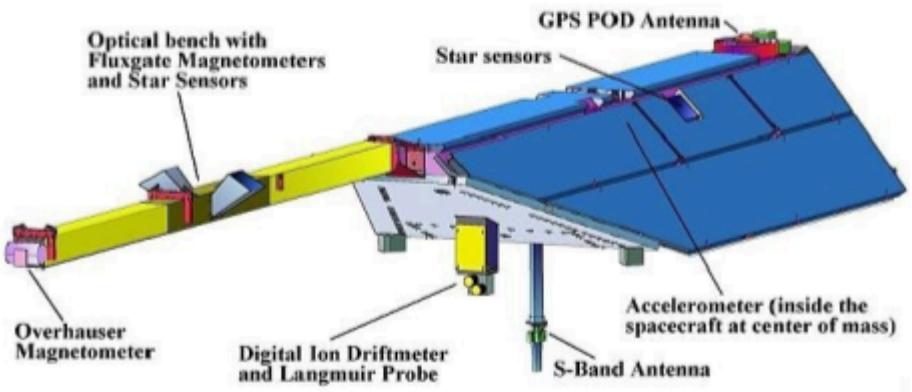
Navigation Orbits



CHAMP High-Low Satellite tracking

- CHAMP: CHallenging Minisatellite Payload
- 2000-2010
- 450km altitude
- Precise Orbit determination with GPS
- Correction for suface forces (drag, solar radiation pressure) with accelerometer
- Applications: static gravity field, magnetic field, GPS occultation





GOCE Gradiometry

- Very low altitude (~250 km)
- 2009-2013
- Gradiometer (3×2 proofmasses)
- Compensation of drag by Ion propulsioin
- High resolution static gravity field, ocean currents

GRACE Low-Low Gravimetry

- Separation ~220km
- 2002-2007
- Inter-satellite Tracking Instrument
  - (~5 $\mu m/s$ in K-band)
- Accelerometer (Surface forces)
- Time-variable gravity field (e.g. Greenland mass loss)

Summary

- For a 2 body problem

    - Central body is point mass
    - Orbit is an ellipse (Kepler orbit)
    1. Orbits are ellipses, with sun (central body) in one of the 2 foci
    2. During equal intervals of time, the connecting line between central body and planet (object) sweeps equal areas
    3. Square of orbital period is proportional to cube of the semi major axis of the ellipse
- Disturbing forces perturb orbit
    - Equatorial bulge of the Earth exerts a moment on the orbit causing it to change orientation in space
- Special orbits, sun synchronous, repeat orbits